

HDI-Predict

Human Development Index Prediction using Machine Learning

Team ID: [Add Team ID]

Repository: <https://github.com/marukurthi/Human-Development-Index>

Program: AICTE–SmartBridge AI/ML Internship (in collaboration with SNU and ServiceNow University)

Overview: This project uses machine learning to predict the Human Development Index (HDI) of a region from key socio-economic indicators — such as income, education, and life expectancy — moving beyond simple GDP-based rankings toward a data-driven, comprehensive measure of well-being.

Team

Name	Role
Chandini Vulli	Team Lead
Kaamisetti Deeven Kumar	Member
Jayasri Thupakula	Member
Nava Jeevan Kumar Chitikena	Member
Jyothi Marukurthi	Member

Tech Stack

Modeling & Analysis	Python, Jupyter Notebook, scikit-learn, pandas, numpy, matplotlib, seaborn
Backend	Python (Flask)
Frontend	HTML5, CSS3
Deployment	Local (not yet deployed to the cloud)

Repository Structure

This documentation follows the project's full development lifecycle, organized into eight phases:

- 1. Brainstorming & Ideation
- 2. Requirement Analysis
- 3. Project Design Phase
- 4. Project Planning Phase
- 5. Project Development Phase
- 6. Project Testing
- 7. Project Documentation
- 8. Project Demonstration

1. Brainstorming & Ideation

Idea Listing

S.No	Idea	Category	Group
1	Predict HDI using GDP, literacy rate, and life expectancy indicators	ML Model	Group 1
2	Compare Linear Regression, Random Forest, and Gradient Boosting models	Model Selection	Group 1
3	Build a web form for users to input custom indicator values	Web App Feature	Group 2
4	Show predicted HDI score instantly after submission	Web App Feature	Group 2
5	Deploy the model as a Flask web application	Deployment	Group 3
6	Add historical HDI trend charts for comparison	Visualization	Group 3

Idea Prioritization

Group	Final Idea	Feasibility	Importance	Status
1	ML-based HDI prediction using GDP, literacy, and life expectancy indicators	High	High	Selected
2	Web form for real-time indicator input and instant HDI prediction	High	High	Selected
3	Historical trend visualization dashboard	Medium	Medium	Future scope

Customer Problem Statement

I am	I'm trying to	But	Because	Feels
A student/researcher analyzing regional development	Estimate the HDI of a region from socio-economic indicators	Doesn't have quick access to a calculated HDI value	Manual HDI computation combines multiple normalized sub-indices and is time-consuming	Frustrated by the lack of a fast, accessible tool

Empathy Map — Persona: Development Data Analyst

SAYS	THINKS
"I need a quick way to estimate HDI without manual calculation"; "I want to compare regions side by side"	Wonders if ML can accurately approximate HDI from raw indicators; questions which indicators matter most
DOES	FEELS
Searches for HDI datasets online; manually calculates HDI in spreadsheets	Overwhelmed by scattered data sources; hopeful automation could save time

2. Requirement Analysis

Functional Requirements

S. No	Category	Description	Priority
1	Authentication	No authentication required; the tool is publicly accessible	Low
2	Authorization levels	Single user role (general user); no admin panel	Low
3	External interfaces	No third-party APIs; uses local Kaggle/UCI CSV datasets only	Medium
4	Transactions processing	Prediction request is processed in real time on form submission	High
5	Reporting	Displays predicted HDI score with a summary of entered indicators	High
6	Business rules	Input indicator values are validated for realistic numeric ranges before prediction	Medium
7	Compliance	No personal data collected; uses only public country-level statistics	Low
8	Other	Model retraining supported via included Jupyter notebooks	Low

Non-Functional Requirements

S. No	Category	Description	Target Metric
1	Performance & Speed	Prediction response returned quickly for a single request	Under 2 seconds
2	Scalability	Designed for individual/local use rather than concurrent traffic	Single-user local deployment
3	Security & Data Privacy	No sensitive personal data stored or transmitted	N/A – public data only
4	Reliability & Availability	Runs reliably on the local Flask server during use	Available during local server uptime
5	Usability & Accessibility	Simple, form-based UI usable by non-technical users	Basic HTML/CSS interface
6	Other	Maintainability via modular Python code (data, model, app layers)	N/A

3. Project Design Phase

Data Flow Diagram

Symbol	Name	In this project
Oval	External Entity	User (enters indicator values through the web form)
Rectangle 1	Process: Validate Input	Checks submitted indicator values are numeric and within realistic ranges
Rectangle 2	Process: Predict HDI	Loads the trained model and generates a predicted HDI score
Solid Rectangle	Data Store	CSV dataset (Kaggle/UCI) and saved model file (pickle)
Arrow	Data Flow	Indicator values → Validate Input → Predict HDI → Predicted Score → User

Solution Architecture

Presentation / Client Layer: Web App (HTML, CSS, Jinja2 templates)

↓↑

API Gateway / Load Balancer: Flask routing (single-server local deployment)

↓↑

Core Logic Service: Data preprocessing & ML prediction (scikit-learn model)

↓↑

Data / Storage Layer: Local CSV dataset and serialized model file (pickle)

Technology Stack

S. No	Layer	Technology	Justification
1	Frontend / Client-Side	HTML, CSS, Jinja2 templates	Simple, lightweight form-based UI served directly by Flask
2	Backend / Server-Side	Python, Flask	Lightweight framework, quick to set up for a single-model ML app
3	Database / Data Storage	CSV files (Kaggle, UCI)	Dataset is static and tabular; no need for a full DBMS at this scale
4	Deployment	Local deployment	Run and demoed locally for the internship submission
5	Version Control	GitHub	https://github.com/marukurthi/Human-Development-Index
6	Libraries	scikit-learn, pandas, numpy, matplotlib, seaborn	Model training, data processing, and visualization

4. Project Planning Phase

Sprint Plan & Product Backlog

Sprint	Epic	Task	Points	Priority	Owner
Sprint -1	Data Collection	Collect and clean the HDI dataset for training	2	High	Team
Sprint -1	Data Collection	Perform exploratory data analysis on the dataset	2	High	Team
Sprint -1	Model Building	Train and compare Linear Regression, Random Forest, and Gradient Boosting models	3	High	Team
Sprint -2	Model Building	Select the best-performing model and save it for reuse	2	High	Team
Sprint -2	Web App	Build a web form for entering socio-economic indicators	2	High	Team
Sprint -2	Web App	Display the predicted HDI score after submission	1	High	Team

Resource Requirements

Resource	Description	Allocation
Computing	CPU specification	Standard CPU (no dedicated GPU required for regression models)
Memory	RAM specification	8 GB
Storage	Disk space for data, models, logs	Local disk (a few hundred MB)
Framework	Python framework	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Dev Environment	IDE	Jupyter Notebook, VS Code
Data	Source, size, format	Kaggle dataset and UCI dataset, CSV format

5. Project Development Phase

Approach

Train and compare Linear Regression, Random Forest, and Gradient Boosting models on HDI-related indicators in Jupyter Notebook, then deploy the best-performing model through a Flask web app. Model training and experimentation live in notebooks; the deployable app (Flask backend + HTML/CSS frontend) is built from the resulting trained model.

Code Quality Checklist

S. No	Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	
2	Proper File Structure	Files and folders are logically organized	Yes	data/, model/, app/
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	
4	Function / Method Names	Functions are descriptively named	Yes	
5	Code Comments	Inline and block comments explain logic	Partial	Can be expanded
6	Modular Design	Code is split into reusable functions/modules	Yes	
7	No Redundant Code	Duplicate or unused code is removed	Yes	
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic handling in app.py

Solution Summary

Repository Link	https://github.com/marukurthi/Human-Development-Index
Programming Language	Python
Frameworks Used	Flask, scikit-learn
Key Features	ML pipeline comparing Linear Regression, Random Forest, and Gradient Boosting; Flask web app for real-time HDI prediction
Pending Features	Historical trend visualization dashboard (future scope)
Setup Instructions	1. Clone the repository 2. pip install -r requirements.txt 3. python app.py 4. Open the local server address in a browser

6. Project Testing

Manual functional testing was carried out against the core prediction workflow and input validation.

Test ID	Description	Input	Expected Result	Status
TC-01	Predict HDI with valid indicator values	GDP, literacy rate, life expectancy within valid ranges	Returns a numeric HDI prediction	Pass
TC-02	Reject invalid/non-numeric input	Text entered in a numeric field	Form shows a validation error	Pass
TC-03	Handle missing input fields	One or more fields left blank	Form prompts the user to fill required fields	Pass
TC-04	Compare predictions against known data	Test rows from the Kaggle/UCI dataset	Predicted values closely match actual HDI values	Pass
TC-05	Repeated submissions	Multiple sequential form submissions	App remains stable, no server errors	Pass

7. Project Documentation

HDI-Predict is a machine learning application that estimates the Human Development Index of a region from key socio-economic indicators (such as income, education, and life expectancy), removing the need to manually combine normalized sub-indices. Linear Regression, Random Forest, and Gradient Boosting models were trained and compared in Jupyter Notebook, and the best-performing model was deployed behind a Flask web application with a simple HTML/CSS form for real-time predictions. The full source, notebooks, and this documentation are available in the project repository.

Repository: <https://github.com/marukurthi/Human-Development-Index>

8. Project Demonstration

Demo video / screenshots: to be added to the repository's Project Demonstration folder.

Live application: not yet deployed — currently runs locally via ``python app.py``.

Repository: <https://github.com/marukurthi/Human-Development-Index>